CA6001 – AI Algorithms Fundamentals & Application

Group Project Technical Report

Machine Learning–Guided Adaptive CPT Site Investigation and Economic Benefit Assessment

Predicting shallow cone-resistance profiles, choosing the next tests, deciding when to stop, and pricing the outcome

Project Title: Machine Learning–Guided Adaptive CPT Site Investigation and Economic Benefit Assessment

Course: CA6001 AI Algorithms Fundamentals & Application

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Student Names and Contributions:

Name	Matriculation no.	Main contributions

Code, configuration and result files: project repository, package `cpt_budget`; figures and tables in this report are regenerated from the saved results.

Summary. Cone penetration tests (CPTs) measure ground resistance along single vertical lines. We test whether AI models can predict unmeasured CPT profiles, choose where to test next and decide when to stop. Five predictors (GP, XGBoost, CNP and two TabPFN hybrids) are replayed on eight sites. Stop rules are frozen on four development sites, and every stop is audited against hidden CPTs. The TabPFN hybrids gave the lowest profile error, within about 3% of XGBoost and GP. No early stop failed the audit (7 early stops in 24 runs), but CPTs were saved only at the most uniform site (21 → 11). Adaptive location choice showed no advantage over a simple space-filling baseline. In the economic scenario, decision delays outweighed CPT savings.

1. Introduction

Domain and motivation. A cone penetration test (CPT) pushes a cone into the ground and records the cone resistance q_c (MPa) with depth. High q_c means stiff ground; low-q_c layers indicate soft material that matters for design. Each test samples one vertical line, so the ground between tests must be inferred. Investigations usually fix the number and positions of CPTs in advance and complete them all, although some add little information. Fewer tests save money, and finishing earlier can bring a whole project forward.

Objective. *With only a few CPTs measured, can an AI model predict the unmeasured profiles well enough to choose the next useful CPTs and to stop once quality is adequate, and what is that worth?* We (i) compare five AI predictors under identical information; (ii) wrap the selected one in an adaptive selection and stopping loop; (iii) audit every stop on hidden CPTs; and (iv) price the resulting CPT counts.

Contribution and scope. The project contributes a replay framework that evaluates prediction, location choice, stopping and economic value under one information boundary (Fig. 1). It reports negative results as well as savings. It uses public CPT data in the 2–8 m depth window. The threshold $q_c < 1$ MPa is a research indicator, not a design criterion, and the economics are counterfactual scenarios.

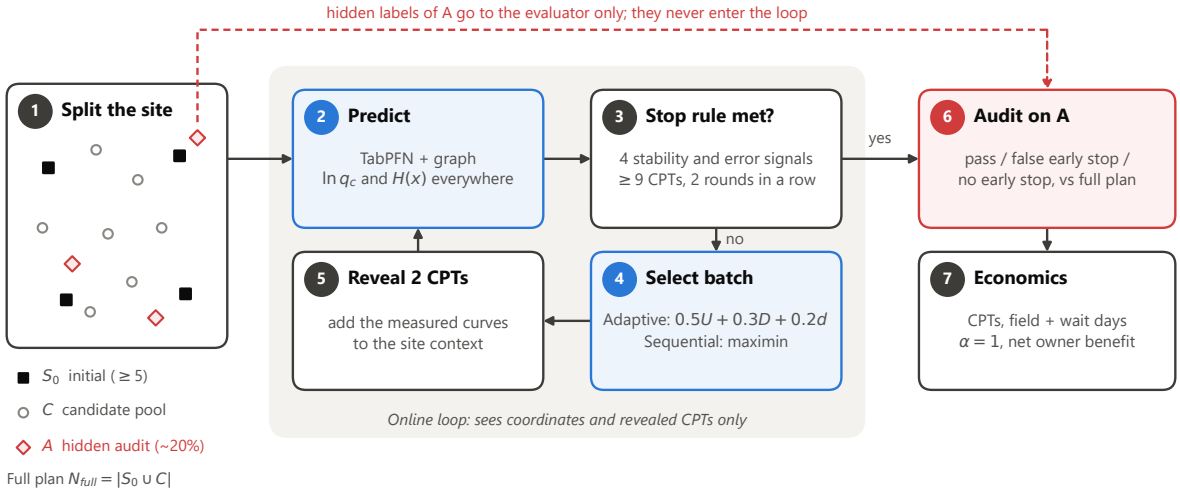


Figure 1: Investigation loop. The hidden audit set A is read only by a separate evaluator after the stop. All policies share the initial set S_0, the candidate pool C and A.

2. Problem Definition

At planned positions x and 24 depth cells z_k (2–8 m, 0.25 m each), the regression target is $y(x, z_k) = \ln(q_c/1\,\text{MPa})$. A *low cell* has $q_c < 1$ MPa ($y < 0$), and the cumulative low-cell thickness is

$$H(x) = 0.25 \sum_{k=1}^{24} \mathbb{1}[y(x, z_k) < 0] \quad \text{(m)}. \tag{1}$$

Each site is split into an initial set S_0, a candidate pool C and a hidden audit set A. Starting from S_0, a policy repeatedly reveals a batch from C and decides whether to stop. It seeks the smallest stopping count N_{stop} that still meets frozen quality targets on A, compared with the full plan of $N_{\text{full}} = |S_0 \cup C|$ CPTs:

$$\min\ N_{\text{stop}} \quad \text{s.t.} \quad \text{log-MAE}_A \le \tau_{\log},\ \ \text{thickness-MAE}_A \le \tau_H,\ \ \text{loss relative to the full plan} \le \varepsilon. \tag{2}$$

The task thus combines *spatial regression* (profiles), *rare-event classification* (low cells), *active learning* (locations) and *optimal stopping*.

Constraints.

- Only coordinates and already revealed curves may enter prediction, selection or stopping.
- Near-duplicate soundings form indivisible groups.
- Sites, not depth cells or seeds, are the replicate unit, so results are descriptive.
- Thresholds are fixed on development sites before evaluation, and a stop that fails the audit stays a failure.

3. Dataset Description

Source. The data are the public ISSMGE TC304 CPT databases (ISSMGE Technical Committee 304, n.d.). They comprise the Global-CPT collection (1,196 soundings from 59 sites; Ching et al., 2023) and the 98 soundings of the New Lock Terneuzen ship lock, operational since August 2025 (Rijkswaterstaat, 2025). Screening leaves 492 usable positions out of 503 raw profiles (495 distinct positions), forming 456 near-duplicate groups at 31 sites. That gives 11,808 depth cells, which are repeated measurements within CPTs, not independent samples. Of the 31 sites, 23 *source* sites only train the models. The 8 *replay* sites, those with at least 20 usable positions, are split into 4 development and 4 evaluation sites.

Features and preprocessing. Each record holds site, local coordinates (m), group, depth and q_c. Raw curves are interpolated to the 24 cell mid-points and q_c is log-transformed. At New Lock, 80 of the 98 files cover the full window. The models' neighbour features are rebuilt from revealed CPTs only, at every round: distances, local medians, low quantiles, low-cell fractions and vertical gradients.

Exploratory analysis. The 0.25 m grid loses some thin layers: its thickness MAE against the raw curves is 0.092 m. Sites differ sharply (Fig. 2). Wellington Airport has no low cells, while New Lock has 24% and strong lateral variability. Across source sites only about 8% of cells are low. Three properties of the data guided the algorithm choice: *few CPTs per site*, *spatial correlation* and *rare low cells*.

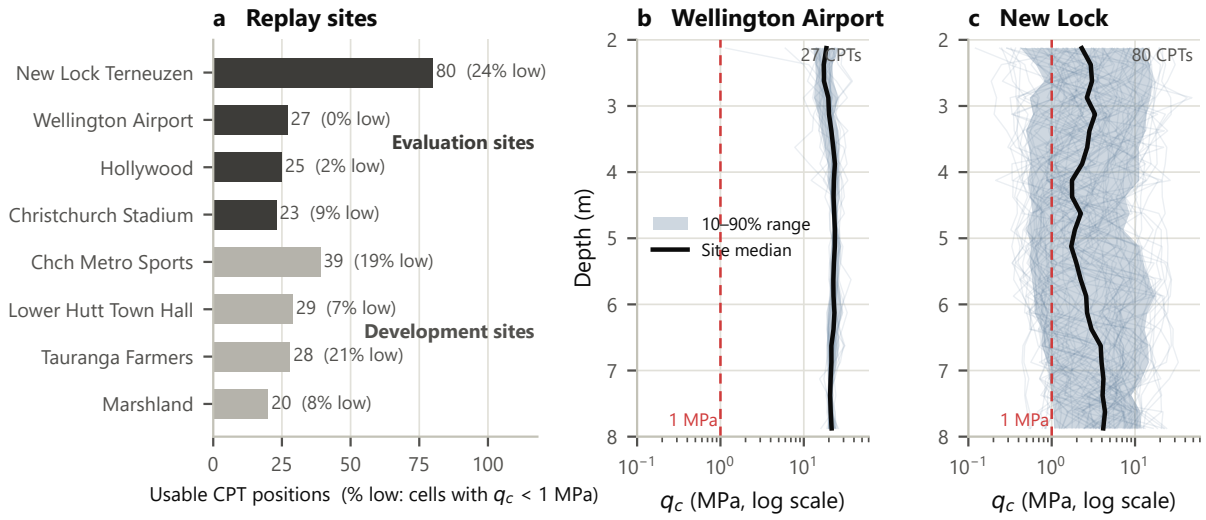


Figure 2: (a) Usable positions and share of low cells at the eight replay sites. (b, c) Profiles at the most uniform and most variable evaluation sites; the band is the 10–90% range and the line the median.

4. Algorithm Selection and Rationale

All five predictors receive the same revealed CPTs at every state. Thickness and low-cell labels are derived from each method's single predicted curve.

Gaussian process (GP). A GP is a prior over functions defined by a kernel. Its prediction is a kernel-weighted combination of the observations, and its variance grows away from them (Rasmussen & Williams, 2006). We use an anisotropic Matérn-3/2 kernel on residuals from a source-site depth trend, refitted each round. It gives native uncertainty, but its smoothness blurs thin low layers.

XGBoost residual. Gradient boosting adds regression trees one at a time, each fitted to the current residuals (Chen & Guestrin, 2016). Here the trees correct an inverse-distance (IDW) estimate using 24 neighbour features. This captures non-linear local effects, but the trees are fixed after source training.

Conditional neural process (CNP). A CNP encodes the revealed profiles relative to the query, pools them in a permutation-invariant way and decodes a full profile (Garnelo et al., 2018). It learns cross-site transfer, but its L1 loss targets the median, so rare low cells are smoothed away.

TabPFN. A Transformer pre-trained on synthetic tabular tasks predicts by in-context learning: the site's revealed rows are passed as context, with no gradient step per round (Hollmann et al., 2025). We fine-tuned it on source sites and then froze it. Two hybrids are tested. *TabPFN joint* blends a TabPFN residual with a triangulation estimate and decides low cells per whole profile. *TabPFN + graph* blends it with a robust graph interpolation and labels a cell low if either branch does, favouring recall.

Compare and contrast. Each method trades off differently:

- GP needs no source data and gives uncertainty, but assumes smooth ground.
- XGBoost and TabPFN exploit cross-site patterns.
- TabPFN also adapts to a new site without retraining.
- CNP is the most data-hungry.

Kriging is close to the GP. Entropy-based sampling (Guan et al., 2022) targets liquefaction, not q_c profiles. Reinforcement learning was rejected because four evaluation sites cannot support a learned policy.

Division of labour. The primary predictor was chosen in advance by the lowest development

log-MAE: TabPFN + graph (0.578), ahead of XGBoost (0.597) and TabPFN joint (0.598). It drives prediction and stopping; the GP supplies uncertainty and XGBoost a second opinion.

5. Methodology

5.1 Evaluation design

The loop is replayed on historical data in a three-level, site-held-out design:

- *Source training*: models are trained on source sites in three folds, with checkpoints chosen on validation sites.
- *Development*: four replay sites select the model, acquisition weights and stop thresholds.
- *Evaluation*: four replay sites (Wellington Airport, Christchurch Stadium, Hollywood, New Lock) test the frozen configuration, each with fold weights that excluded it.

For each site and seed (17, 29, 43), whole groups are assigned to A (≈20%), a dispersed S_0 (≥5 positions) and C. Three policies share each split:

- **Full plan** (B_{full}) completes all of $S_0 \cup C$.
- **Sequential** (B_{seq}) adds the two candidates farthest from any revealed CPT, a space-filling *maximin* baseline.
- **Adaptive** (P_{adapt}) adds the two candidates with the highest acquisition score.

Both Sequential and Adaptive use the same stop rule. Comparing them isolates the value of location choice from that of stopping early. Evaluation gives $4 \times 3 \times 2 = 24$ trajectories plus 12 full-plan runs.

Metrics (all on A):

- log-MAE, the mean $|\hat{y} - y|$;
- thickness MAE, the mean $|\hat{H} - H|$;
- low-cell precision, recall, F1 and FPR.

Averages are taken within seed, then site, then equally across sites.

5.2 Acquisition score and stop rule

For each candidate group the adaptive policy computes three signals:

- U, the mean GP standard deviation;
- D, the share of cells where the primary model and XGBoost disagree on low cells;
- d, the distance to the nearest revealed CPT.

It then ranks each signal within the pool and scores

$$a(x) = 0.5\,\text{rank}(U) + 0.3\,\text{rank}(D) + 0.2\,\text{rank}(d). \tag{3}$$

The weights put uncertainty first and use spacing as a tie-breaker. The one alternative tried, which drops D (0.7, 0, 0.3), lost on development.

Each round the controller computes four signals over all planned positions:

- ΔQ_{90}, the 90th percentile of the change in predicted $\ln q_c$;
- ΔH_{90}, the same for thickness;
- a leave-group-out log-MAE on revealed CPTs;
- the 90th percentile of the GP standard deviation.

It stops when all four pass for two consecutive rounds, after at least nine CPTs and full quadrant coverage. The earliest possible stop is therefore 11 CPTs. Thresholds are the 0.50 quantile of each signal at acceptable development states, chosen to maximise verified saving minus false-stop rate; the 0.75 and 0.90 quantiles gave 17% and 58% false stops. A stop *passes the*

audit if log-MAE ≤ 0.626, thickness MAE ≤ 0.881 m and neither is worse than the full plan by more than 0.05 or 0.25 m. Low-cell recall ≥ 0.5 with FPR ≤ 0.1 is a separate diagnostic.

Table 1: Hyperparameters. Model settings were chosen on source validation sites, decision settings on development sites; none on evaluation sites.

Component	Setting
GP	Matérn $\nu = 3/2$, length scales from (20, 20, 1) m, white noise; refitted by marginal likelihood
XGBoost	3-NN IDW base + $w\times$trees; depth 6, learning rate 0.3, early stopping (36–37 trees); w per fold
CNP	64 hidden units, mean + max pooling, L1 loss; 26,817 parameters; 500–1,200 steps
TabPFN	v2 weights, fine-tuned with CRPS + MSE for 500–600 updates; joint blend 0.5/0.5
Graph	4 neighbours, vertical weight 0.15, pseudo-Huber $\delta = 0.35$
Stop rule	$\Delta Q_{90} \leq 0.082$, $\Delta H_{90} \leq 0$ m, support log-MAE ≤ 0.584, GP-std P90 ≤ 0.461

5.3 Economic model and tools

The owner's net benefit relative to the full plan is

$$B_{\text{net}} = (C_{\text{full}} - C_{\text{policy}}) + \alpha\, \Delta T_{\text{inv}}\, (c_{\text{day}} + b_{\text{day}}) - C_{\text{AI}}. \tag{4}$$

The terms are:

- C: €750 per CPT;
- ΔT_{inv}: calendar days saved, at 5 CPTs per rig-day plus a 0.25-day decision delay per batch;
- c_{day} = €5,000 avoidable project cost per day;
- b_{day} = €3,000 earlier-opening benefit per day;
- C_{AI} = €600.

The main scenario sets $\alpha = 1$ (every day saved shortens the project); 0.5 and 0.75 are sensitivities. Only the remobilisation and review rates come from a 2025 Dutch quotation (MOS Grondmechanica B.V., 2025); the rest are stated assumptions (U.S. Government Accountability Office, 2015, 2020), applied to every site.

Tools: Python 3.11, scikit-learn 1.9 (Pedregosa et al., 2011), XGBoost 3.2, PyTorch 2.11, TabPFN 9.2.0 (Prior Labs, n.d.) and matplotlib, on an RTX 4080 GPU.

6. Implementation

The code is a Python package (`cpt_budget`) whose modules mirror the information boundary:

- data splits;
- models with a common `condition(revealed)`/`predict(query)` interface, which strips queries of measured values;
- acquisition and stop rule;
- replay loop;
- a separate audit evaluator;
- economics.

The frozen configuration stores SHA-256 fingerprints of data, weights and settings. The core of batch selection is:

```
while remaining and chosen_count < batch_size:
    dist  = [min_distance(g, observed_xy) for g in remaining]
    score = 0.5*group.rank_U + 0.3*group.rank_D + 0.2*rank01(dist)
    gid   = argmax(score)                          # ties: group id, then location id
    chosen += members(gid)                         # whole near-duplicate group
```

```
    observed_xy = vstack([observed_xy, xy(gid)])  # coordinates only, never curves
```

Each round is logged; for example, the adaptive run at Wellington Airport (seed 17) stops in round 3:

```
round n_obs dQ90  dH90 support_logMAE gp_std_P90 streak stop
  3    11   0.067 0.00     0.132        0.169      2    online_stability
```

Key decisions. TabPFN weights stay frozen, so each round only updates the context and takes seconds. Grouping is enforced throughout. All 14 CNP and 129 TabPFN weight tensors differ from initialisation, so weak CNP results are not an untrained-model artefact.

Checks. Automated checks show that audit labels cannot affect predictions (24 paired contexts, difference 0). They also confirm that group counts are exact, inputs are unchanged and economic sums recompute. All 95 unit tests pass.

7. Results and Evaluation

7.1 Prediction quality

Table 2: Prediction quality (equal-site means). †Primary model. Bold: best of the five main methods; CNP's 0% FPR is excluded because it never predicts a low cell. Recall excludes Wellington Airport, which has no low cells.

	Development (4 sites)		Evaluation (4 sites)				
Method	log-MAE	Thick. (m)	log-MAE	Thick. (m)	Recall (%)	FPR (%)	F1
GP	0.658	0.614	0.509	**0.400**	14.0	**1.9**	0.11
XGBoost residual	0.597	0.571	0.512	0.485	31.2	6.9	0.30
CNP	0.833	0.774	0.657	0.430	0.0	0.0	0.00
TabPFN joint	0.598	**0.483**	**0.497**	0.491	33.3	7.8	0.30
TabPFN + graph†	**0.578**	0.586	0.503	0.595	**36.3**	9.6	**0.31**
TabPFN ablation (the fine-tuned joint row above completes the set)							
Residual, pretrained weights	0.627	0.642	0.506	0.494	23.4	5.9	0.23
Residual, fine-tuned weights	0.598	0.626	0.502	0.509	22.4	6.0	0.22
Residual, fine-tuned + sign guard	0.598	0.573	0.503	0.497	32.1	7.3	0.30
Joint, pretrained weights	0.604	0.485	0.495	0.480	34.9	7.5	0.32

Comparison. TabPFN joint has the lowest evaluation profile error, 0.497 (Table 2). That is 2.9% below XGBoost; TabPFN + graph is 1.8% below. The margins are small: joint beats XGBoost in 9 of 12 site–seed cells, and at New Lock the GP is better than both hybrids (0.976 against 1.034). The GP has the lowest evaluation thickness error (0.400 m), mainly because it rarely predicts low cells (recall 14%). The graph union gives the highest recall (36%) at the highest FPR (9.6%). CNP never predicts a low cell. Site difficulty dominates (Fig. 3): log-MAE is 0.11–0.24 at Wellington Airport but 0.98–1.14 at New Lock.

Ablation and error analysis. Fine-tuning improved the raw residual on development sites (0.627 → 0.598). On evaluation sites, however, pretrained and fine-tuned joint models differ by under 0.002, and most of the thickness gain comes from the spatial blend (0.626 → 0.483 m). The improvement therefore belongs to the hybrid, not to TabPFN alone. At New Lock, error peaks at 4–6 m, where 23% of cells are low: the primary model finds 53% of them, with 85 false alarms, while the GP flags only 1–3% of cells against 9–23% true.

7.2 Selection, stopping and audit

The adaptive score samples differently from maximin (Fig. 4). At New Lock it concentrates on the variable central cluster, whereas maximin spreads out. At the three easier sites audit quality is inside the targets from the first few CPTs (Fig. 5). At New Lock it plateaus above both targets, even with all 64 CPTs.

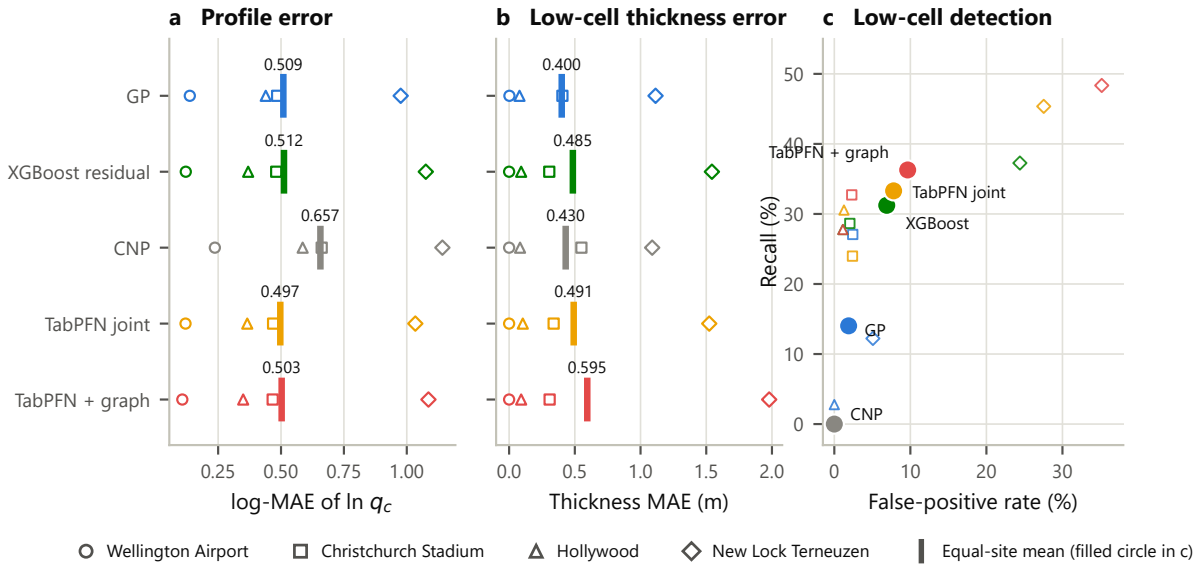


Figure 3: Five methods at the four evaluation sites. Open marks: site means (shape = site); bars and filled circles: equal-site means. (c) Recall against FPR.

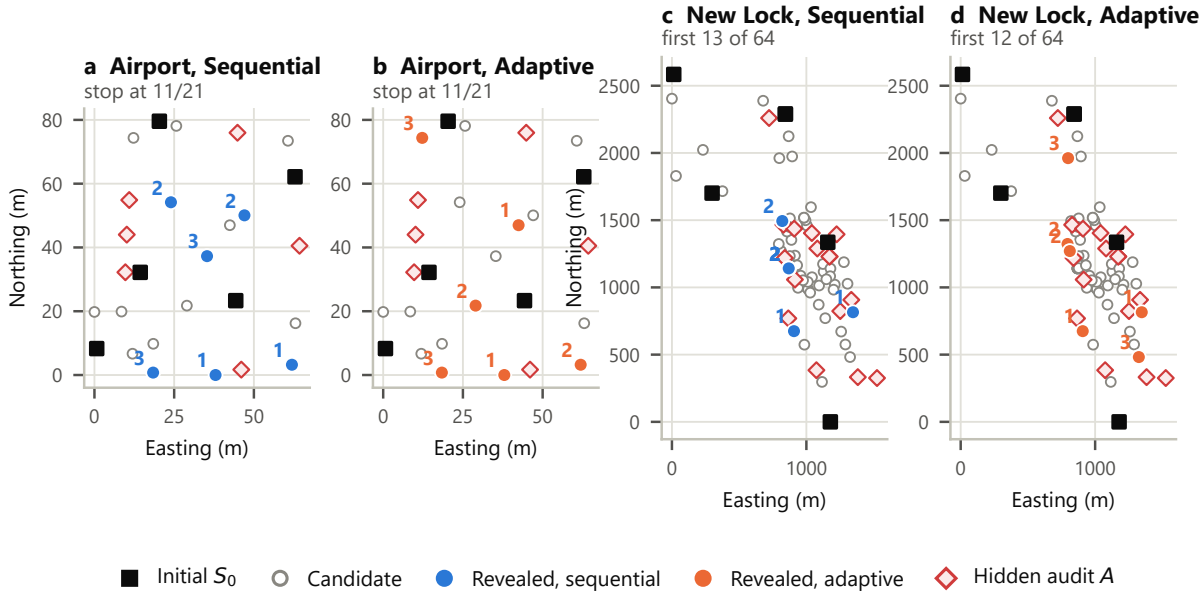


Figure 4: Selection maps, seed 17: Wellington Airport at the stop (a, b) and New Lock after 12–13 CPTs (c, d). Numbers give the round in which a position was revealed.

Table 3: Decision outcomes at the evaluation sites (means of three seeds). $|A|$: audit positions. "Full plan passes": seeds in which even the full plan meets the targets. Oracle N: smallest passing count, found with hindsight. Net: owner's benefit of the adaptive policy at $\alpha = 1$.

			CPTs at stop		Saving (%)		Full plan	Oracle	Net
Site	N_{full}	$\lvert A\rvert$	Seq.	Adapt.	Seq.	Adapt.	passes	N (adapt.)	k€
Wellington Airport	21	6	11.0	11.0	47.6	47.6	3/3	9	+23.3
Christchurch Stadium	18	5	18.0	18.0	0.0	0.0	3/3	9	−12.6
Hollywood	20	5	18.7	20.0	6.7	0.0	3/3	13–14	−11.9
New Lock	64	16	64.0	64.0	0.0	0.0	0/3	none	−55.9
Equal-site mean					13.6	11.9			−14.3

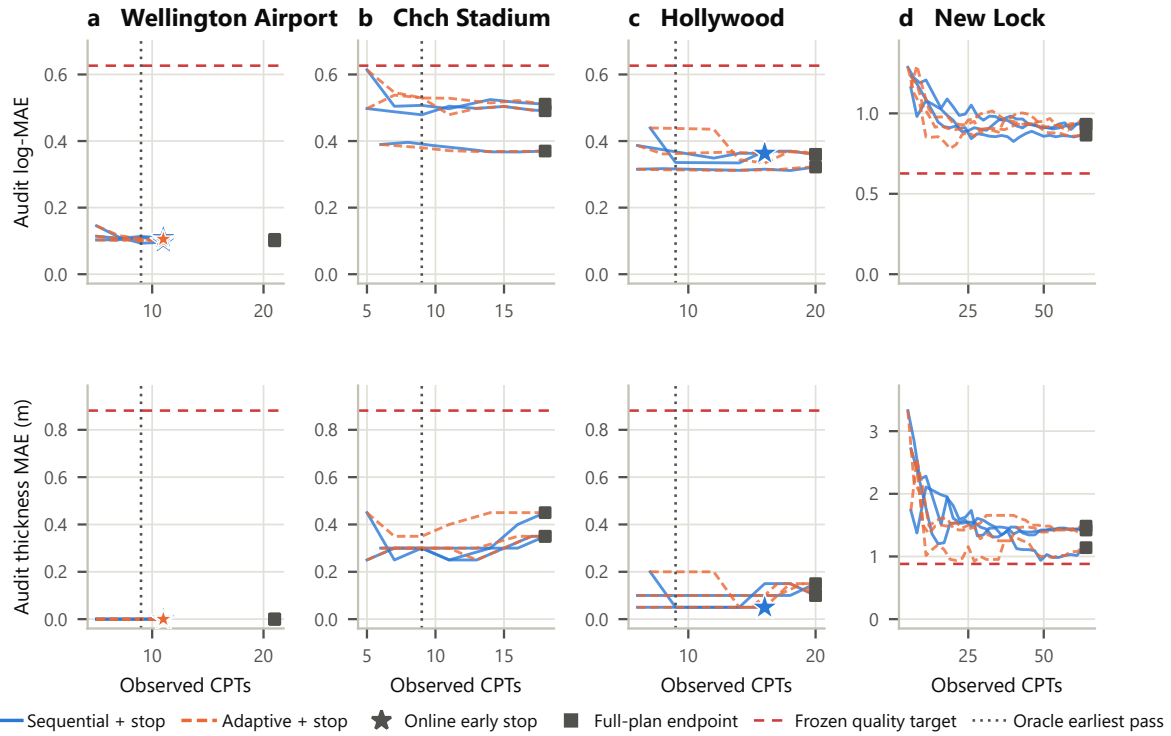


Figure 5: Hidden-audit quality as CPTs are added (three seeds per policy). Stars: online early stops; squares: full plan; red: frozen targets; dotted: oracle earliest passing count.

Stopping outcomes. No early stop failed the audit (Table 3), but the evidence is thin. Only 7 of 24 evaluation runs stopped early: 6 at Wellington Airport, where every state passes, and one at Hollywood. The rule's real value is that it never stopped at New Lock, where any early stop would have failed. By contrast, a naive stop at the ninth CPT makes 7 false stops. The leave-group-out support signal carries this safety: without it, 2–3 false stops appear.

Savings. Equal-site savings were 13.6% (Sequential) and 11.9% (Adaptive). Both come almost entirely from Wellington Airport (21 → 11 CPTs), which is the earliest stop the rule allows. Adaptive selection showed no detectable advantage: across all 24 site–seed pairs, its stopping count differed from Sequential only 3 times, in both directions.

A conservative rule. With hindsight, Christchurch Stadium met the targets with 9 of 18 CPTs and Hollywood with 13–14 of 20, yet both adaptive runs went to the full plan. Several signals block jointly. $\Delta H_{90} \leq 0$ still allows 10% of positions to change, and its zero value is a quantile tie, not a designed tolerance. GP uncertainty tracked audit error only weakly (Spearman $\rho = 0.21$ against 0.60 on development sites). Low-cell targets failed at all three sites with low cells, and audit sets are small (5–16 positions).

7.3 Economic value

At Wellington Airport the adaptive policy yields +€23,300 (Fig. 6). This comprises:

- €7,500 from ten avoided CPTs;
- €16,400 from finishing 2.05 days earlier;
- minus €600 of AI cost.

Sequential earns the same, so the gain comes from staged stopping, not from AI location choice.

The other sites lose €11,900–€55,900 despite using no extra CPTs, because each batch adds a 0.25-day decision delay (6.9 days over 27–29 transitions at New Lock). With zero delay, those losses shrink to the €600 AI cost. With a one-day delay, New Lock would lose €222,000. The

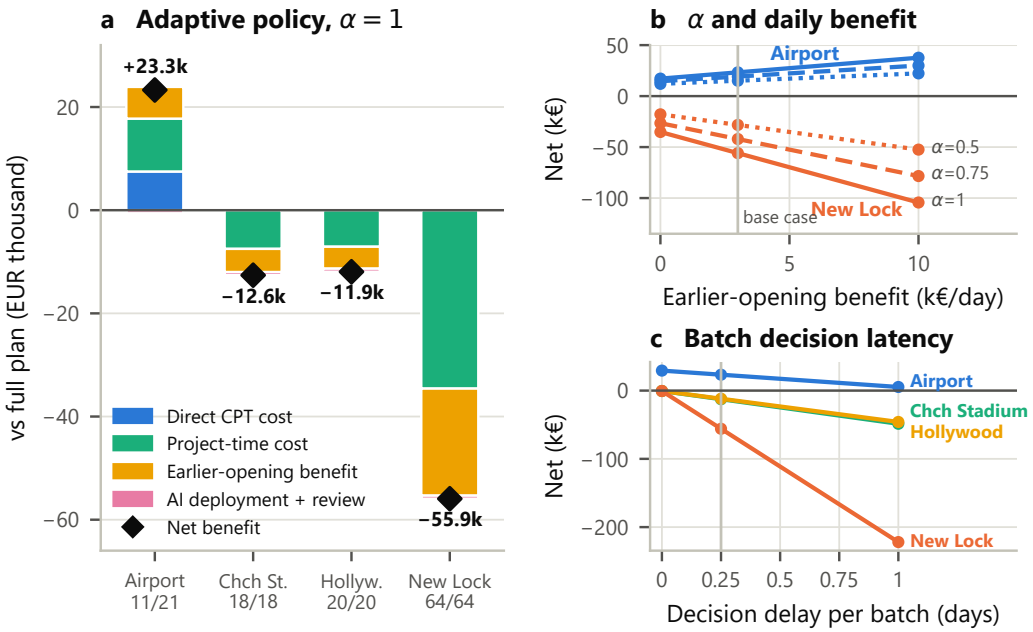


Figure 6: Owner's financial scenario, adaptive policy (mean of three seeds). (a) Components at $\alpha = 1$. (b) Sensitivity to α and the daily opening benefit. (c) Sensitivity to the decision delay per batch.

four-site mean is −€14,300 (Adaptive) and −€13,500 (Sequential). No site changes sign within any tested sensitivity range.

8. Discussion and Insights

What worked. The pipeline is auditable: hidden labels never reach the controller, every stop is audited, and failures are reported. The stop rule vetoed every stop at the one site where stopping would have failed. **What the results reveal.**

1. *Site predictability decided the savings, more than the algorithm did.* Savings appeared only where five CPTs already describe the ground.
2. *Location choice added nothing measurable.* Staged investigation plus a stop rule produced the benefit.
3. *Models matter through their decisions.* Profile errors lie within 3%, but low-cell behaviour differs sharply (GP cautious, graph sensitive, CNP blind), which changes what a stop rule can safely conclude.
4. *Time dominates the economics.* Decisions must return within the working day, and unpredictable sites should revert to the full plan early.

Threats to validity.

- The development sites were checkpoint-selection sites during source training. This likely explains why learned models beat the GP by 12% there but by 1% on evaluation sites, and it means the model ranking is within noise.
- The evaluation sites had been inspected in an earlier project stage, so this is a re-evaluation, not a blind test.
- With 4 sites, 5–16 audit positions per site and 7 early stops, the 95% upper bound on the false-stop rate is about 40%.
- New Lock is the only non-New-Zealand site and the largest, so domain shift and heterogeneity are confounded there.
- The economics rest on assumed values and on $\alpha = 1$.

Future improvements. Quality-predictive stop signals would take the clear opportunities at

Christchurch Stadium and Hollywood. A site-predictability check after the initial CPTs would catch New Lock-like sites early. Same-day decisions would remove most of the delay cost. An event-weighted loss on a 0.1 m grid would help low cells. Finally, the method should be tested on genuinely new sites with real cost data.

9. Conclusion

We built and audited an end-to-end AI-guided CPT investigation loop combining five predictors, an uncertainty–disagreement–distance acquisition score, a multi-signal stop rule and an economic model. The TabPFN hybrids predicted best, but only by about 3%. The frozen stop rule never stopped wrongly, yet it saved CPTs only at the most uniform site (47.6%, +€23,300 in the main scenario), and adaptive selection did not beat maximin. The main lesson is that an AI algorithm should be judged by the decision it supports, with an independent audit and honest negative results. Here, site predictability and decision latency mattered more than predictor accuracy.

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